



## SRI LANKA TECHNOLOGY CAMPUS

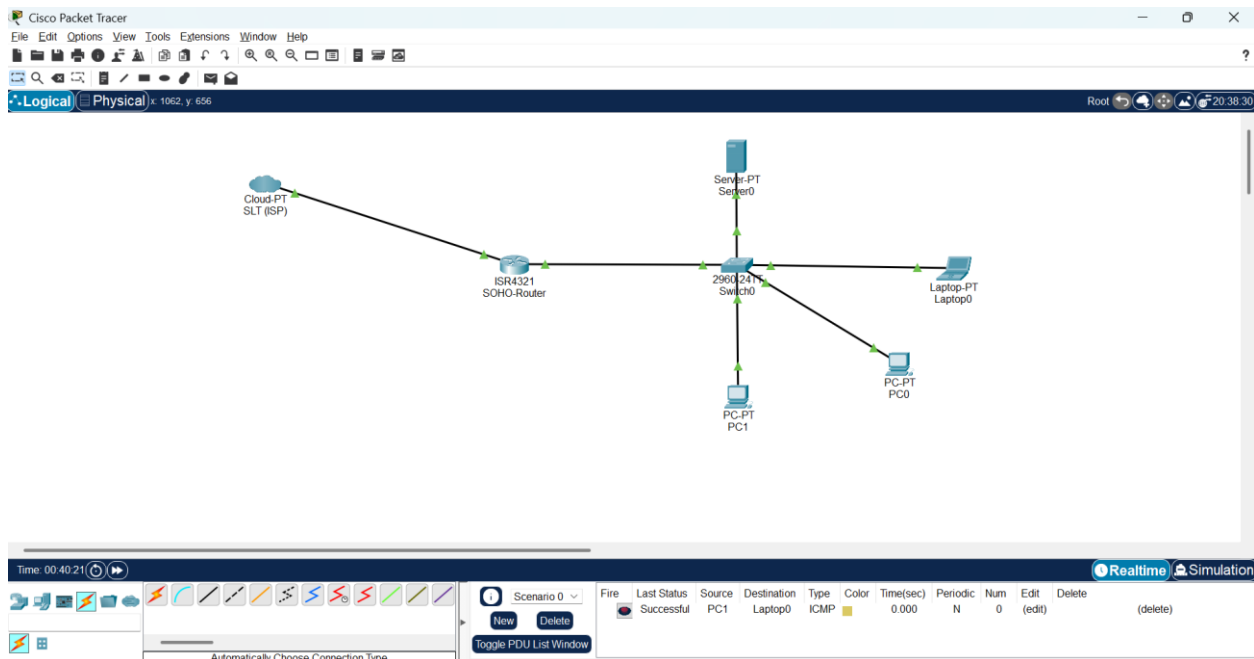
### ASSIGNMENT 02

### MINI PROJECT

**Module: Internet Technologies (CIT201)**

**Student ID:23DA2-0268**

**Topic: Implementing a SOHO Network with Services**



### Step 1, SOHA Router Configuration

```
SOHO-Router
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if) #
Router(config-if) #
Router(config-if) #
Router(config-if) #
Router(config-if) #
Router(config-if) #
Router(config-if) #hostname SOHO-Router
SOHO-Router(config) #
SOHO-Router(config) #
SOHO-Router(config) #interface g0/0/0
SOHO-Router(config-if) #ip address 192.168.10.1 255.255.255.0
SOHO-Router(config-if) #no shutdown

SOHO-Router(config-if) #
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

SOHO-Router(config-if) #exit
SOHO-Router(config) #
SOHO-Router(config) #interface g0/0/1
SOHO-Router(config-if) #ip address 203.0.113.1 255.255.255.0
SOHO-Router(config-if) #no shutdown

SOHO-Router(config-if) #
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

SOHO-Router(config-if) #exit
SOHO-Router(config) #
SOHO-Router(config) #interface g0/0/0
SOHO-Router(config-if) #ip nat inside
SOHO-Router(config-if) #exit
SOHO-Router(config) #interface g0/0/1
SOHO-Router(config-if) #ip nat outside
SOHO-Router(config-if) #exit
SOHO-Router(config) #access-list 1 permit 192.168.10.0 0.0.0.255
SOHO-Router(config) #ip nat inside source list 1 interface g0/0/1 overload
SOHO-Router(config) #
SOHO-Router(config) #ip domain-name myoffice.com
SOHO-Router(config) #crypto key generate rsa
The name for the keys will be: SOHO-Router.myoffice.com
Choose the size of the key modulus in the range of 360 to 4096 for your

Copy Paste
```

## Step 2, Configure NAT

```
SOHO-Router(config)#  
SOHO-Router(config)#interface g0/0/0  
SOHO-Router(config-if)#ip nat inside  
SOHO-Router(config-if)#exit  
SOHO-Router(config)#interface g0/0/1  
SOHO-Router(config-if)#ip nat outside  
SOHO-Router(config-if)#exit  
SOHO-Router(config)#access-list 1 permit 192.168.10.0 0.0.0.255  
SOHO-Router(config)#ip nat inside source list 1 interface g0/0/1 overload  
SOHO-Router(config)#  
SOHO-Router(config)#
```

## Step 3, Bonus Task (SSH Security)

```
SOHO-Router(config)#  
SOHO-Router(config)#  
SOHO-Router(config)#ip domain-name myoffice.com  
SOHO-Router(config)#crypto key generate rsa  
The name for the keys will be: SOHO-Router.myoffice.com  
Choose the size of the key modulus in the range of 360 to 4096 for your  
General Purpose Keys. Choosing a key modulus greater than 512 may take  
a few minutes.  
  
How many bits in the modulus [512]: 1024  
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]  
  
SOHO-Router(config)#username admin secret cisco123  
*Mar 1 0:11:25.994: %SSH-5-ENABLED: SSH 1.99 has been enabled  
SOHO-Router(config)#line vty 0 4  
SOHO-Router(config-line)#transport input ssh  
SOHO-Router(config-line)#login local  
SOHO-Router(config-line)#exit  
SOHO-Router(config)#
```

## Step 4, DHCP

Server0

Physical Config **Services** Desktop Programming Attributes

**SERVICES**

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

**DHCP**

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.10.1

DNS Server: 192.168.10.2

Start IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Maximum Number of Users: 246

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

| Pool Name  | Default Gateway | DNS Server   | Start IP Address | Subnet Mask   | Max User | TFTP Server | WLC Address |
|------------|-----------------|--------------|------------------|---------------|----------|-------------|-------------|
| serverPool | 192.168.10.1    | 192.168.10.2 | 192.168.10.10    | 255.255.255.0 | 246      | 0.0.0.0     | 0.0.0.0     |

## Step 5, DNS

The screenshot shows the 'Server0' configuration window with the 'Services' tab selected. On the left, a 'SERVICES' list includes HTTP, DHCP, DHCPv6, TFTP, DNS (highlighted), SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, and Radius EAP. The main area is titled 'DNS' and shows 'DNS Service' as 'On'. Under 'Resource Records', a record is configured with 'Name' 'www.myoffice.com', 'Type' 'A Record', and 'Address' '192.168.10.2'. Below this are 'Add', 'Save', and 'Remove' buttons. A table lists the configured records:

| No. | Name             | Type     | Detail       |
|-----|------------------|----------|--------------|
| 0   | www.myoffice.com | A Record | 192.168.10.2 |

## Step 6, Laptop Web browsers

The screenshot shows the 'Laptop0' configuration window with the 'Desktop' tab selected. A 'Web Browser' window is open, displaying the URL 'http://www.myoffice.com'. The page content includes:

### My SOHO Network Portal

Welcome to the Office Intranet

This secure server provides internal services to authorized personnel only. If you can see this page, your DHCP and DNS configuration is working correctly!

**Department Quick Links:**

- [IT Documentation & Policies](#)
- [Company Copyrights Info](#)
- [Network Topology Diagram](#)
- [View Company Logo](#)

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Thank You!